Muting Durably Reduces Engagement with Untrustworthy Online Sources; Media Literacy Doesn't

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Pre-registered: OSF

Existing approaches

Media literacy (cognitive)

- Tips to improve discernment (Pennycook et al. 2021; Guess et al. 2020; Hoes et al. 2024; Maertens et al. 2021)

Exposure change (environmental)

- Platform deactivation (Allcott et al. 2024); algorithmic intervention (Guess et al. 2023); reducing exposure to like-minded sources (Nyhan et al. 2023) or to untrustworthy sources (Bergeron-Boutin et al. 2026)
- Increasing exposure to "good" content (Bail et al. 2018; Levy 2021)

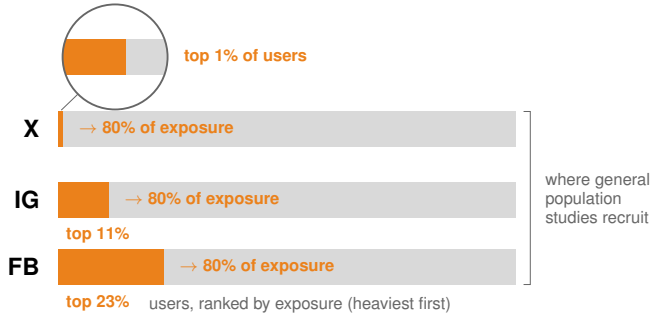
Existing approaches: studying the wrong crowd?

Media literacy (cognitive)

- Tips to improve discernment (Pennycook et al. 2021; Guess et al. 2020; Hoes et al. 2024; Maertens et al. 2021)

Exposure change (environmental)

- Platform deactivation (Allcott et al. 2024); algorithmic intervention (Guess et al. 2023); reducing exposure to like-minded sources (Nyhan et al. 2023) or to untrustworthy sources (Bergeron-Boutin et al. 2026)
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A small minority accounts for ∼80% of exposure (Grinberg et al. 2019; Bergeron-Boutin et al. 2026) — but most interventions are tested on the general population.

Research question

How can we reduce engagement with untrustworthy sources among **the people who actually consume them**?

Media literacy

improve how users *evaluate* the information they encounter

vs.

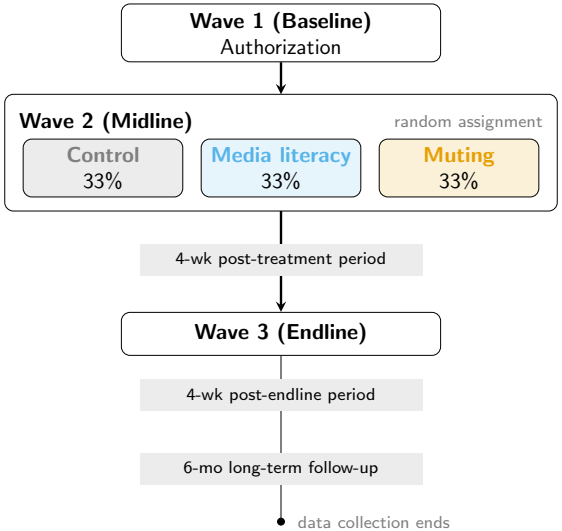
Muting

reduce *exposure* to problematic sources

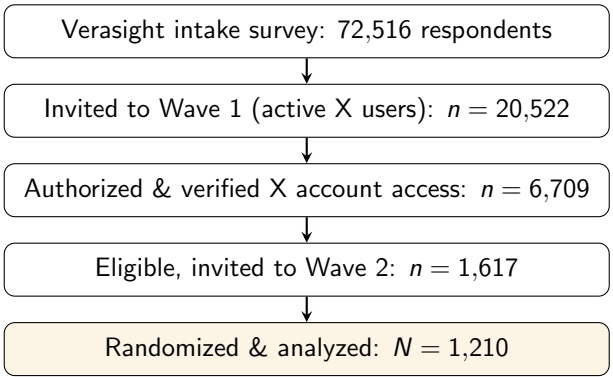
Population: Active X users who actually engaged with untrustworthy sources

Horizon: effects tracked long-term (+6 months)

Study design: Three-arm RCT on X



Recruiting a hard-to-reach population



28.3% reported being active X users

51% completed authorization

$\sim$24% of authorized users were eligible

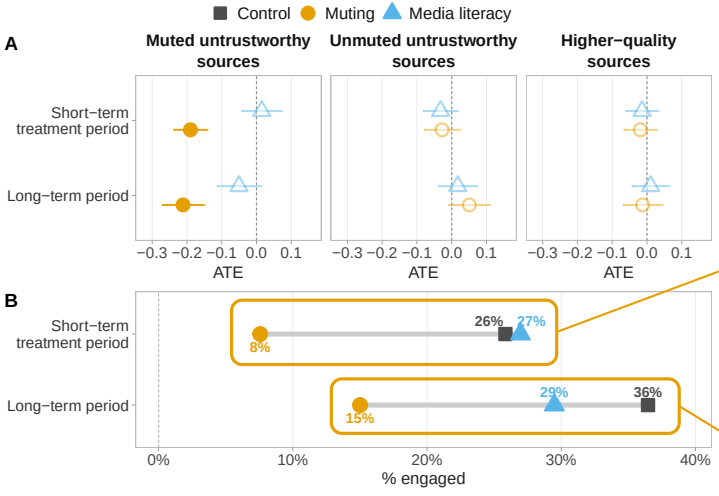
Treatment arms

	Muting ($P = 1/3$)	**Media lit.** ($P = 1/3$)	**Control** ($P = 1/3$)
Muting	Keep randomly sampled 304 untrustworthy sources muted for 4 weeks ($15)	No muting	No muting
Tips	Placebo tips + 3 weekly reminders	Media literacy tips + 3 weekly reminders	Placebo tips + 3 weekly reminders

Outcomes

- **Engagement** (behavioral, from X API):
  - Separately for muted vs. unmuted untrustworthy sources, and higher-quality sources
- **Discernment** (survey):
  - Sharing and accuracy discernment (true vs. false headlines)
- **Attitudes** (survey):
  - Trust in untrustworthy sources
  - Willingness-to-accept for muting
  - Confidence in vote count; affective polarization

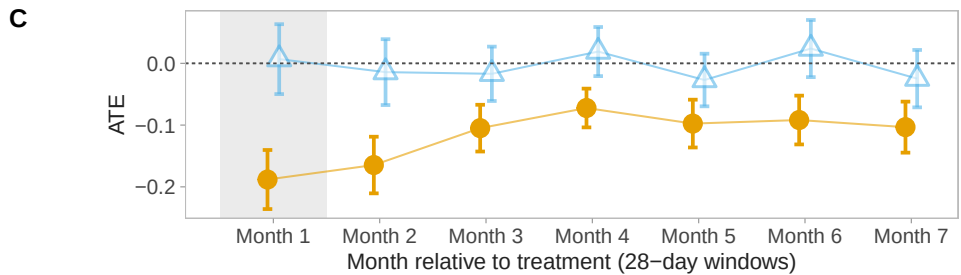
Muting durably reduces engagement with muted sources



19 pp reduction in any engagement (26%→8%; ∼69% decline); $p < .001$, $q = .001$

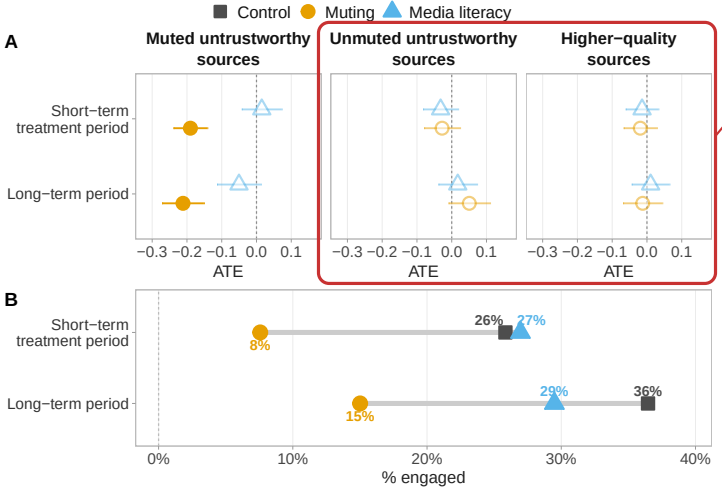
21 pp reduction in any engagement (36%→15%; ∼58% relative reduction); $p < .001$, $q = .001$

Trajectory of muting effects on muted sources over time



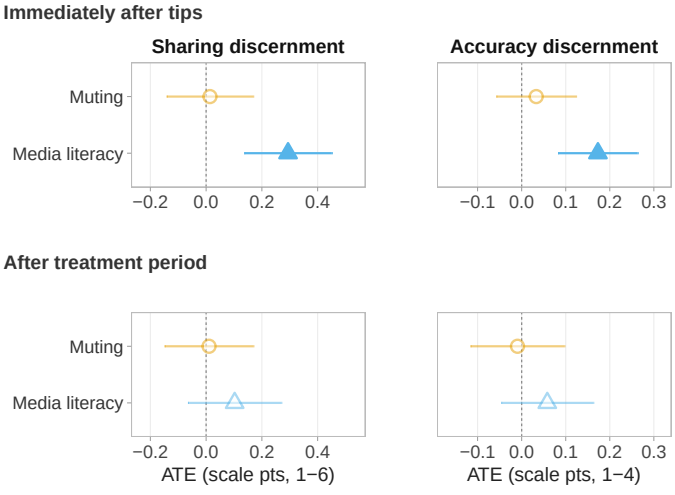
- Muted sources (any engagement): largest at Month 1 (-19 pp, $p < .001$), remains significant ($p < .001$) through Month 7 (-10 pp)

No substitution



No substitution to unmuted or high-quality sources

Media literacy improves discernment in Wave 2—but fades by Wave 3



Media literacy improved sharing/accuracy discernment in W2 ($q = .002$), but effects gone by W3 despite 3 weekly reminder emails

Key findings

1. Muting durably reduces engagement with muted sources
   - Effects persist for up to six months: engagement with muted sources falls 58% relative to control, and 87% remained fully compliant — all 304 sources still muted — 6+ months later
   - No substitution to unmuted untrustworthy or higher-quality sources
2. Muting reduces trust in muted sources
3. Media literacy has no significant effect on engagement behaviors, despite producing temporary improvements in discernment

Structural changes that limit exposure to untrustworthy sources are more effective at changing behavior than cognitive interventions among those with low-quality information diets.

Implications

1. **Consumption is environment-dependent**
   Driven by visibility, not lack of discernment (junk food analogy: reducing access > nutrition labels)
2. **No "whack-a-mole" problem**
   Users don't substitute to other untrustworthy sources when muted sources disappear
3. **Low-effort, opt-in tools can produce durable change**
   High compliance from a single low-effort action — effects persist without continued incentives
4. **Multiple implementation paths**
   Platform features, third-party services (e.g., Bluesky moderation lists), or voluntary user adoption

Questions?



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THE MERCURY PROJECT

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Appendix

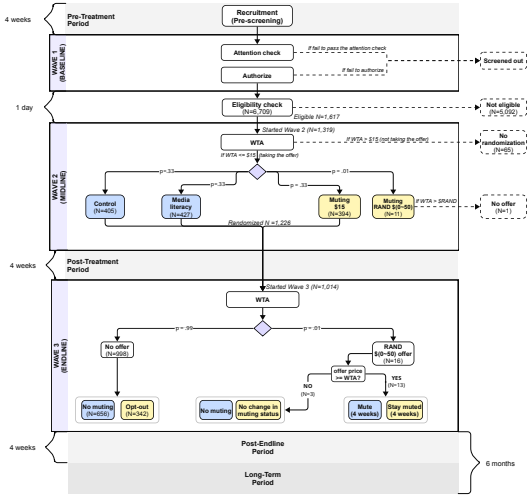
How does our study differ from prior approaches?

Prior approach	Gap / Difference from our study
Media literacy (cognitive)	
Tips to improve discernment (Pennycook et al. 2021; Guess et al. 2020; Hoes et al. 2024; Maertens et al. 2021)	Mixed evidence on long-term persistence
Exposure change (environmental)	
Platform deactivation (Allcott et al. 2024)†	Bundled; reduces exposure to good and bad content alike; not source-specific
Algorithmic intervention (Guess et al. 2023)†	Requires platform cooperation; not user-driven
Reducing exposure to like-minded sources (Nyhan et al. 2023)†; increasing exposure to cross-partisan content (Bail et al. 2018; Levy 2021)	Trying to solve different problems (polarization/echo chambers); + increasing good $\neq$ decreasing bad
Reducing exposure to untrustworthy sources (Bergeron-Boutin et al. 2026)†	**Closest approach**: oversampled high-exposure users, cut exposure $\sim$70% — but tested on attitudes only, never on (long-term) engagement

Our study goes further, targeting **engaged** users (not just exposed), reduces **engagement** (not just exposure), for up to **6 months**

† Part of the 2020 US Facebook/Instagram Election Study; academic–industry collaboration requiring direct platform cooperation.

Full participant flow diagram



Recruitment pipeline: Detailed numbers

- 72,516 respondents recruited through Verasight's intake survey
- 28.3% ($n = 20{,}522$) reported being active X users $\rightarrow$ invited to Wave 1
- Among those, 51% completed X account authorization successfully ($n = 6{,}709$)
- Among authorized users: $\sim$24% eligible (account $\geq$4 mo. old; engaged with $\geq$1 untrustworthy source) $\rightarrow$ invited to Wave 2 ($n = 1{,}617$)
- 1,226 Wave 2 completers took the WTA offer and were randomized; excluding 16 participants who received the incentive-compatibility random offer in Wave 3 yields the final analytic sample: $N = 1{,}210$

Challenges:

- Lower X usage rates than pilot (declining platform usage)
- Required multiple recruitment adjustments mid-study
- Embedded screening questions directly into Verasight's intake survey
- Deployed abbreviated Wave 1 for failed authorizations
- Weekly eligibility re-checks (Jan–June 2025) surfaced newly eligible participants

Design pivot: Unfollowing → muting

- Original plan: reduce exposure by **unfollowing** untrustworthy accounts
- Platform API changes during fielding made unfollowing infeasible

- Pivoted to **muting**, which offered several advantages:
  - No need to know if participant follows an account
  - Affects "For You" algorithmic feed (not just chronological)
  - No change to public profile information (follower lists)
  - Lower visibility to participants (less reactance)

- Muting reduces visibility of content from muted sources, **except:**
  - Directly visiting a muted account's profile
  - Indirect exposure from a third party (e.g., a friend retweets a non-muted user's reply to a muted account)
  - Quoted posts from muted sources (warning label shown, but still viewable)

- **Lesson:** Platform instability requires flexible research designs; interventions are constrained by what platforms permit

Which accounts were muted?

- Combined two third-party lists of untrustworthy sources:
  - **NewsGuard:** rating < 60 (U.S., English, active)
  - **Top FIBers:** FIB Index > 10 (DeVerna et al. 2024)
- Filtered out inactive accounts, non-English accounts, and political actors
- Kept accounts responsible for 95% of follows in pilot ($= 434$ accounts)
- For each participant: randomly muted 70% of the treatment list ($= 304$ accounts), stratified by pilot engagement decile
- The other $\sim$1,104 untrustworthy accounts in the full inventory ($1{,}408 - 304$; $\sim$78%) remain unmuted $\rightarrow$ allows testing for substitution
- Muting randomized independently at the participant level

Balance at randomization (Wave 2, midline)

Covariate	Control	Muting	Diff	p	Control	Media Lit.	Diff	p
Democrat	0.41	0.44	0.03	0.37	0.41	0.46	0.04	0.22
Republican	0.48	0.46	−0.02	0.58	0.48	0.42	−0.06	0.10
Non-white	0.25	0.28	0.03	0.38	0.25	0.29	0.04	0.20
Age	42.49	42.53	0.04	0.97	42.49	42.56	0.07	0.94
College grad	0.40	0.35	−0.05	0.14	0.40	0.40	0.00	0.99
Male	0.53	0.49	−0.04	0.27	0.53	0.48	−0.05	0.14
Engagement (low)	5.07	4.85	−0.22	0.68	5.07	4.53	−0.54	0.27
Engagement (high)	0.91	0.77	−0.14	0.31	0.91	1.03	0.12	0.49
True mean acc.	2.92	2.86	−0.05	0.26	2.92	2.87	−0.05	0.32
False mean acc.	1.64	1.68	0.04	0.37	1.64	1.65	0.01	0.85
True mean sharing	5.81	5.92	0.11	0.24	5.81	5.79	−0.02	0.81
False mean sharing	5.31	5.32	0.01	0.95	5.31	5.30	−0.01	0.93
Sharing discernment	0.50	0.60	0.11	0.24	0.50	0.48	−0.02	0.87
Accuracy discernment	1.28	1.18	−0.09	0.12	1.28	1.22	−0.06	0.34
# obs.	402	389			402	419		
F-stat	1.08				0.74			
p-value	0.37				0.76			

Unadjusted two-sided *t*-tests per covariate; joint *F*-test from regressing treatment on all covariates (missing-indicator adjustment).

Muting-vs-Media-Literacy columns omitted for space.

Balance at endline (Wave 3)

Covariate	Control	Muting	Diff	p	Control	Media Lit.	Diff	p
Democrat	0.39	0.44	0.05	0.17	0.39	0.46	0.07	0.06
Republican	0.48	0.45	−0.03	0.45	0.48	0.42	−0.06	0.14
Non-white	0.25	0.28	0.03	0.37	0.25	0.28	0.04	0.29
Age	42.25	42.61	0.36	0.74	42.25	43.07	0.82	0.44
College grad	0.40	0.36	−0.04	0.29	0.40	0.40	0.00	0.93
Male	0.54	0.49	−0.05	0.20	0.54	0.47	−0.07	0.08
Engagement (low)	5.31	4.92	−0.39	0.51	5.31	4.55	−0.76	0.18
Engagement (high)	0.91	0.83	−0.08	0.58	0.91	0.98	0.07	0.72
True mean acc.	2.92	2.85	−0.06	0.22	2.92	2.88	−0.04	0.44
False mean acc.	1.63	1.67	0.04	0.37	1.63	1.63	0.01	0.91
True mean sharing	5.79	5.90	0.10	0.32	5.79	5.72	−0.07	0.49
False mean sharing	5.32	5.29	−0.03	0.77	5.32	5.29	−0.03	0.78
Sharing discernment	0.48	0.61	0.13	0.18	0.48	0.43	−0.04	0.68
Accuracy discernment	1.29	1.18	−0.11	0.10	1.29	1.24	−0.05	0.49
# obs.	317	342			317	339		
F-stat	1.11				1.01			
p-value	0.34				0.45			

Covariate balance among participants remaining at the Wave 3 endline survey. Unadjusted two-sided t-tests per covariate; joint F-test from regressing treatment on all covariates (missing-indicator adjustment). Muting-vs-Media-Literacy columns omitted for space.

Attrition: Survey-based (Wave 2 → Wave 3)

Wave 2 randomized group	Attrited (%)	Stayed (%)
Control	21.1% (85)	78.9% (317)
Media literacy	19.1% (80)	80.9% (339)
Muting	12.1% (47)	87.9% (342)

$\chi^2(2) = 12.331$, $p = 0.002$, Cramer's $V = 0.092$ — muting attrited less than control and media literacy

Comparison	N	Diff. (pp)	95% CI	t	df	p (raw)	p (BH)
Muting − Control	791	−9.1	[−14.2, −3.9]	−3.45	762	0.001	0.002
Muting − Media Literacy	808	−7.0	[−12.0, −2.0]	−2.76	796	0.006	0.009
Media Literacy − Control	821	−2.1	[−7.6, 3.4]	−0.73	814	0.464	0.464

Welch's two-sample t-tests on the binary attrition indicator; Benjamini–Hochberg adjusted p-values, as preregistered.

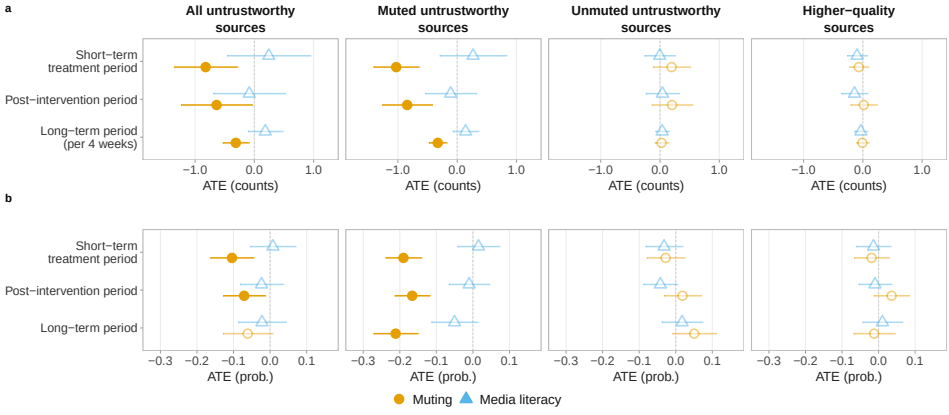
Attrition: Engagement-data availability

Wave 2 randomized group	Post-endline (4-week) Attrited (%)	Post-endline (4-week) Stayed (%)	Long-term (6-month) Attrited (%)	Long-term (6-month) Stayed (%)
Control	7.5% (30)	92.5% (372)	10.0% (40)	90.0% (362)
Media literacy	4.8% (20)	95.2% (399)	6.9% (29)	93.1% (390)
Muting	4.9% (19)	95.1% (370)	9.3% (36)	90.7% (353)

Unlike survey-based attrition, engagement-data availability was **balanced** across arms

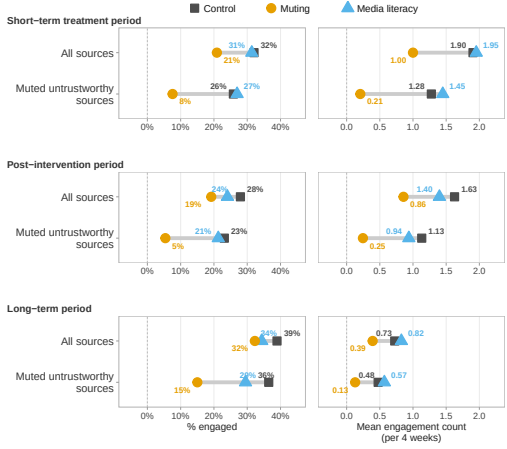
Post-endline: $\chi^2(2) = 3.473$, $p = .176$, $V = .035$ | Long-term: $\chi^2(2) = 2.616$, $p = .270$, $V = .023$

Treatment effects on engagement with untrustworthy sources



OLS, HC2 SEs. Post-treatment = 4 wks after W2; post-endline = 4 wks after W3; long-term = 6 mo. after W3 (rescaled to 4-wk equiv.). Opaque = $q < .05$ (BKY FDR). Panel (a) total engagement count; panel (b) any engagement.

Distribution shift and descriptive means



Raw proportion engaging at least once (left) and raw mean engagement count (right), for all and muted untrustworthy sources, across the three periods.

Treatment effects on engagement (full table)

DV	Term	Estimate	SE	p-value	q-value
Total engagement	Muting	−0.821	0.270	0.0024	0.002
	Media literacy	0.243	0.358	0.4963	0.248
	Muting vs. ML	−1.065	0.319	0.0008	0.001
Binary engagement	Muting	−0.104	0.030	0.0005	0.001
	Media literacy	0.008	0.031	0.8037	0.366
	Muting vs. ML	−0.112	0.029	0.0001	0.001
Total eng. × muted	Muting	−1.029	0.194	<.0001	0.001
	Media literacy	0.268	0.287	0.3495	0.184
	Muting vs. ML	−1.297	0.249	<.0001	0.001
Binary eng. × muted	Muting	−0.190	0.025	<.0001	0.001
	Media literacy	0.015	0.029	0.6045	0.282
	Muting vs. ML	−0.205	0.025	<.0001	0.001

HC2 robust SEs. Unadjusted p-values and sharpened FDR q-values. All outcomes measured during the four-week post-treatment (incentive) period.

Engagement with unmuted untrustworthy sources

DV	Term	Estimate	SE	p-value	q-value
Total eng. × unmuted	Muting	0.195	0.160	0.2212	0.787
	Media literacy	−0.005	0.130	0.9719	0.787
	Muting vs. ML	0.200	0.159	0.2082	0.787
Binary eng. × unmuted	Muting	−0.028	0.026	0.2936	0.787
	Media literacy	−0.032	0.026	0.2135	0.787
	Muting vs. ML	0.004	0.025	0.8606	0.787

No evidence of behavioral backlash or compensatory engagement with unmuted untrustworthy sources

Engagement with high-quality sources

DV	Term	Estimate	SE	p-value	q-value
Total eng. × HQ	Muting	−0.069	0.080	0.3929	1.000
	Media literacy	−0.098	0.085	0.2472	1.000
	Muting vs. ML	0.029	0.072	0.6877	1.000
Binary eng. × HQ	Muting	−0.019	0.024	0.4322	1.000
	Media literacy	−0.014	0.024	0.5577	1.000
	Muting vs. ML	−0.005	0.024	0.8302	1.000

Untrustworthy and trustworthy information are not substitutes

Discernment results (full table)

DV	Term	Estimate	SE	p-value	q-value
Sharing discernment (W2)	Muting	0.014	0.079	0.8613	0.702
	Media literacy	0.294	0.080	0.0003	0.002
	Muting vs. ML	−0.280	0.078	0.0003	0.002
Accuracy discernment (W2)	Muting	0.033	0.046	0.4739	0.702
	Media literacy	0.173	0.046	0.0002	0.002
	Muting vs. ML	−0.140	0.047	0.0027	0.007
Sharing discernment (W3)	Muting	0.010	0.081	0.8973	0.702
	Media literacy	0.102	0.085	0.2306	0.380
	Muting vs. ML	−0.092	0.083	0.2693	0.380
Accuracy discernment (W3)	Muting	−0.009	0.054	0.8621	0.702
	Media literacy	0.058	0.053	0.2749	0.380
	Muting vs. ML	−0.068	0.051	0.1882	0.380

HC2 robust SEs. Unadjusted p-values and sharpened FDR q-values. Lee bounds for W3 (accounting for differential attrition): sharing $[-0.251, 0.228]$; accuracy $[-0.198, 0.129]$ — both include zero.

Trust and demand (full table)

DV	Term	Estimate	SE	p-value	q-value
Trust in untrust. sources	Muting	−0.093	0.046	0.0431	0.125
	Media literacy	−0.057	0.047	0.2280	0.250
	Muting vs. ML	−0.036	0.045	0.4215	0.457
Trust in muted sources	Muting	−0.169	0.056	0.0027	0.015
	Media literacy	−0.098	0.056	0.0799	0.153
	Muting vs. ML	−0.072	0.054	0.1827	0.238
Trust in unmuted sources	Muting	−0.026	0.054	0.6322	0.457
	Media literacy	−0.026	0.056	0.6381	0.457
	Muting vs. ML	0.000	0.053	0.9934	0.520
WTA for untrust. sources	Muting	1.387	1.011	0.1702	0.238
	Media literacy	−2.041	1.013	0.0442	0.125
	Muting vs. ML	3.428	0.887	0.0001	0.002

HC2 robust SEs. WTA winsorized at 95th percentile. Lee bounds: trust in muted sources $[-0.420, -0.043]$; WTA (muting) $[-1.838, 2.943]$; WTA (muting vs. ML) $[0.787, 4.762]$.

Post-endline engagement (full table)

DV	Term	Estimate	SE	p-value	q-value
Total engagement	Muting	−0.637	0.306	0.0375	0.046
	Media literacy	−0.086	0.310	0.7811	0.353
	Muting vs. ML	−0.551	0.282	0.0509	0.054
Binary engagement	Muting	−0.071	0.029	0.0155	0.026
	Media literacy	−0.023	0.030	0.4351	0.240
	Muting vs. ML	−0.048	0.028	0.0853	0.081
Total eng. × muted	Muting	−0.844	0.214	<.0001	0.001
	Media literacy	−0.109	0.219	0.6172	0.328
	Muting vs. ML	−0.735	0.192	0.0001	0.001
Binary eng. × muted	Muting	−0.166	0.024	<.0001	0.001
	Media literacy	−0.010	0.028	0.7159	0.352
	Muting vs. ML	−0.156	0.023	<.0001	0.001

HC2 robust SEs. Unadjusted p-values and sharpened FDR q-values. No evidence of spillover to unmuted or higher-quality sources (see Fig. 1).

Long-term engagement (6 months post-Wave 3)

HC2 robust SEs,

DV	Term	Estimate	SE	p-value	q-value
Total engagement	Muting	−0.312	0.112	0.0052	0.007
	Media literacy	0.184	0.148	0.2120	0.146
	Muting vs. ML	−0.497	0.131	0.0002	0.001
Binary engagement	Muting	−0.061	0.035	0.0766	0.071
	Media literacy	−0.022	0.033	0.5064	0.217
	Muting vs. ML	−0.039	0.033	0.2403	0.151
Total eng. × muted	Muting	−0.328	0.076	<.0001	0.001
	Media literacy	0.142	0.110	0.1998	0.146
	Muting vs. ML	−0.469	0.100	<.0001	0.001
Binary eng. × muted	Muting	−0.211	0.031	<.0001	0.001
	Media literacy	−0.050	0.032	0.1185	0.098
	Muting vs. ML	−0.161	0.029	<.0001	0.001

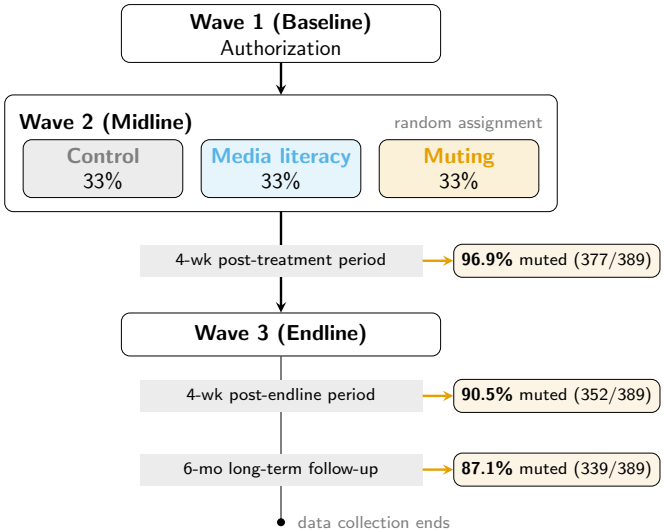
relative to control. No evidence of spillover to unmuted or higher-quality sources at long-term follow-up (see Fig. 1).

Political attitudes (full table)

DV	Term	Estimate	SE	p-value	q-value
Confidence in vote count	Muting	−0.125	0.052	0.0166	0.111
	Media literacy	−0.084	0.054	0.1198	0.214
	Muting vs. ML	−0.041	0.057	0.4796	0.268
Feelings towards outparty	Muting	−0.270	1.794	0.8805	0.416
	Media literacy	2.760	1.871	0.1406	0.214
	Muting vs. ML	−3.030	1.822	0.0964	0.214

No significant effects on political attitudes after FDR adjustment

High compliance that persists for 6+ months



Participants rarely chose to reverse the intervention, even though they could unmute at any time—even 6+ months later

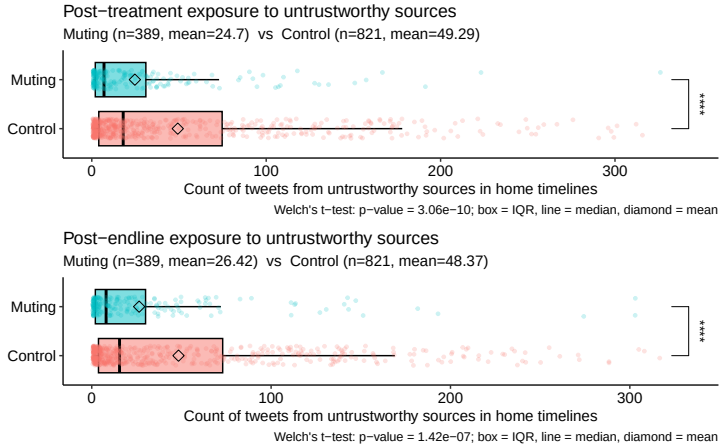
% and n/N are of Wave 2 completers ($N = 389$) still muted at each check.

Manipulation check: Compliance across all time points

		W2 completers ($N = 389$)		W3 completers ($N = 342$)	
Time point	Status	N	%	N	%
Post-treatment	Complied	377	96.9	–	–
	Non-compliance	10	2.6	–	–
	Muting failed	2	0.5	–	–
Post-endline	Complied	352	90.5	314	91.8
	Non-compliance	32	8.2	28	8.2
	No data / server issue	4	1.0	–	–
Long-term (6+ months)	Complied	339	87.1	302	88.3
	Non-compliance	45	11.6	40	11.7
	No data / server issue	4	1.0	–	–

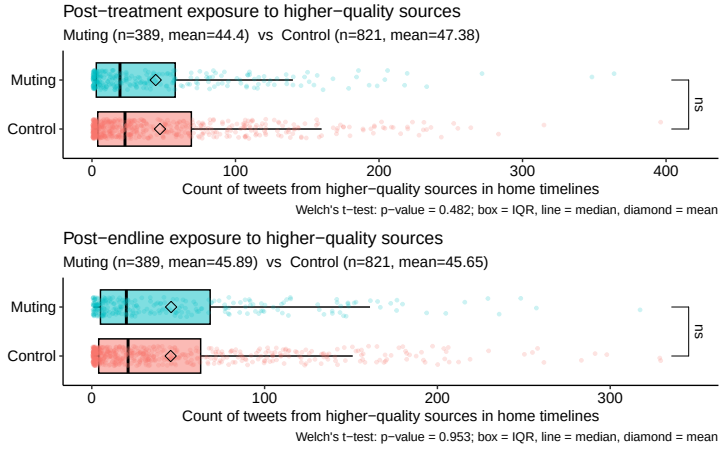
Long-term compliance check conducted Feb 24–25, 2026, >6 months after last W3 completions.

Muting worked: exposure to untrustworthy sources fell



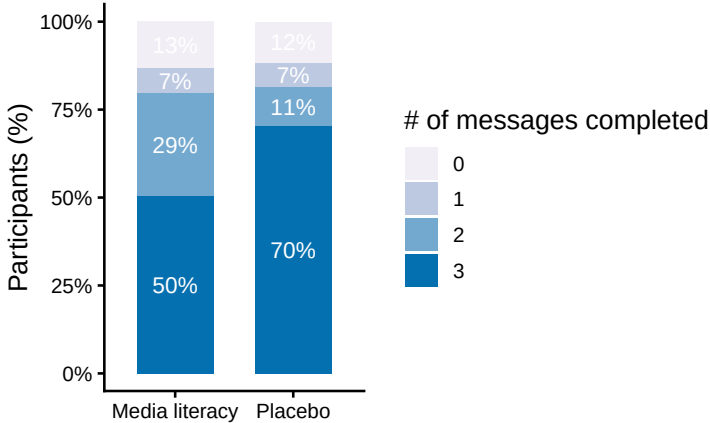
Potential exposure: tweets from untrustworthy sources appearing in participants' home timelines (via X API), muting vs. non-muting (control + media literacy combined), post-treatment and post-endline periods; Wave 2 completers.

...without changing exposure to higher-quality sources



Exposure to higher-quality-source tweets in home timelines: no significant difference between muting and non-muting conditions at either period — muting selectively reduced exposure to untrustworthy sources only.

Manipulation check: Media literacy reminder compliance



- Compliance: completing the Qualtrics survey linked at the end of each of 3 weekly reminder emails
- 87% of media literacy participants (88% placebo) completed at least one reminder

Alternative explanations?

Muting operating as a reversible *barrier to exposure* that most participants accepted and did not undo

Could the decline instead reflect...	**Evidence**
Reduced demand for untrustworthy content?	**No**, muting did not lower WTA to keep muting
Broader disengagement from X?	**No**, no differences in total engagement on X across conditions
A data-collection artifact (API caps)?	**No**, very few observations capped, none in the long-term follow-up

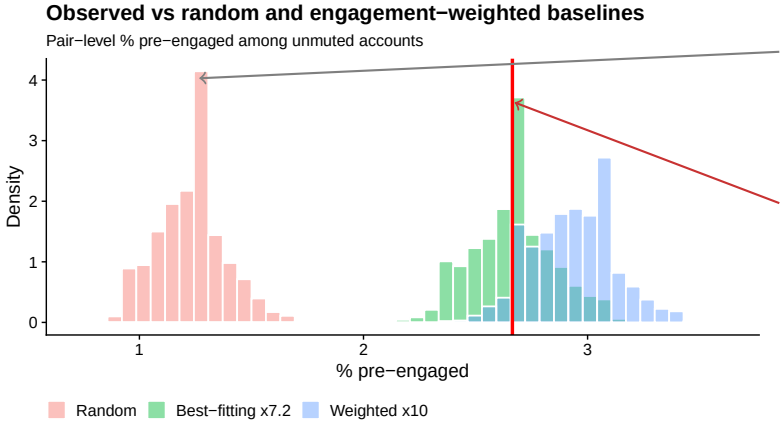
Mechanism: Self-reported exposure

DV	Term	Estimate	SE	p-value	q-value
Muted untrust. sources	Muting	−0.298	0.059	<.0001	0.001
	Media literacy	0.039	0.064	0.5488	0.260
	Muting vs. ML	−0.336	0.062	<.0001	0.001
Unmuted untrust. sources	Muting	−0.187	0.062	0.0027	0.005
	Media literacy	0.056	0.062	0.3695	0.189
	Muting vs. ML	−0.243	0.062	<.0001	0.001
Higher-quality sources	Muting	−0.125	0.063	0.0468	0.033
	Media literacy	0.027	0.061	0.6591	0.282
	Muting vs. ML	−0.152	0.061	0.0128	0.013

Wave 3 self-reported exposure (1–5 scale) to muted, unmuted, and higher-quality sources, regressed on Wave 2 randomized group. HC2 robust SEs.

N: muted (985), unmuted (981), higher-quality (991).

Mechanism: Selective, not random unmuting

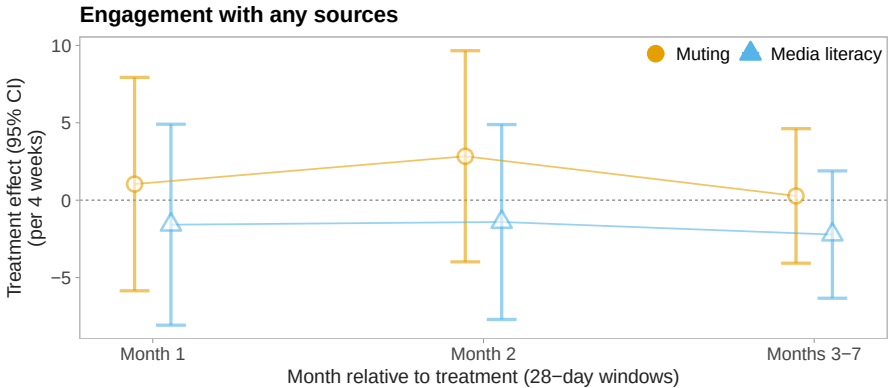


If reversal were passive, unmuting would look **random** ($\sim$1.2% pre-engaged)

Observed rate (2.66%, red line) matches a $\sim$7.2$\times$ engagement-weighted model (green)

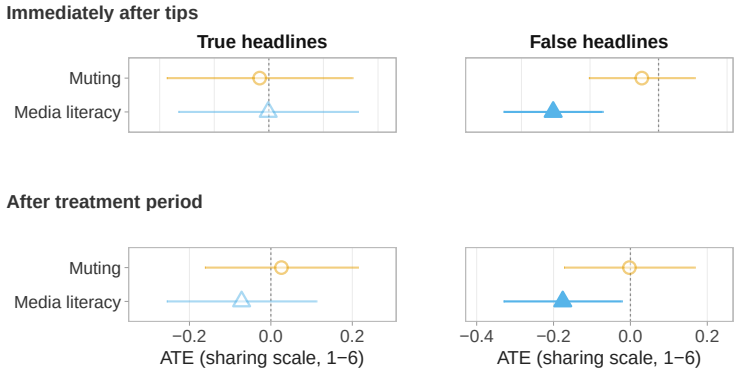
Deliberate, active decision, not passive forgetting

Mechanism: No broader platform disengagement



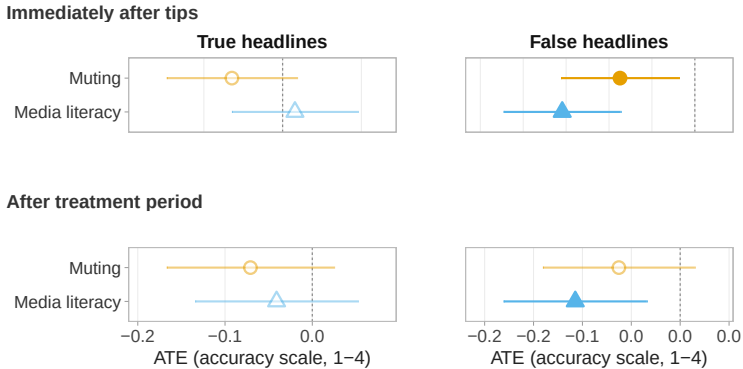
- Participants did not simply use X less overall.

Discernment decomposition: Sharing intentions



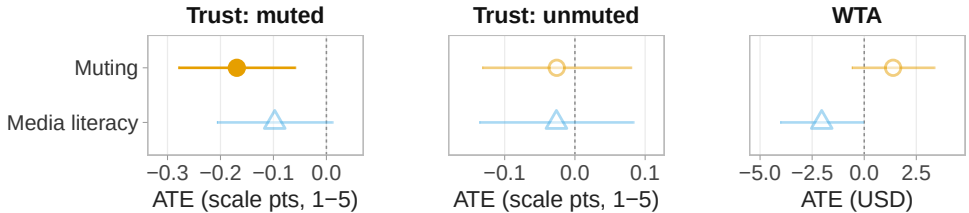
Sharing intention for true vs. false headlines, Waves 2 and 3. Unadjusted $p < .05$ (filled = significant); not FDR-corrected.

Discernment decomposition: Accuracy ratings

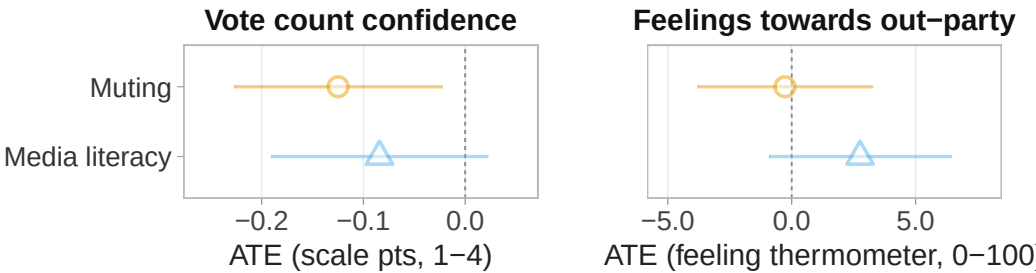


Accuracy rating for true vs. false headlines, Waves 2 and 3. Unadjusted $p < .05$ (filled = significant); not FDR-corrected.

Trust and WTA coefficient plots



Political attitudes coefficient plots



Estimation strategy

- **ITT:** $Y_i = \alpha + \tau_1 \text{Muting}_i + \tau_2 \text{MediaLit}_i + \rho X_i + \varepsilon_i$
- Covariates X_i selected via lasso (pre-registered)
- HC2 robust standard errors
- Two-sided tests; sharpened FDR q-values alongside unadjusted p-values
- Pairwise contrasts (muting vs. ML) via `glht()` in R
- Pre-registered on OSF (October 4, 2024) prior to main study launch

Media literacy treatment: Tips provided

Participants received 7 tips for evaluating online news:

1. Be skeptical of sensational headlines
2. Check the source
3. Look for unusual formatting
4. Evaluate photos critically
5. Consider the motivations of the source
6. Distinguish satire from real news
7. Examine website domains

- 3 weekly reminder emails at days 7, 14, and 21 after Wave 2
- Comprehension/confirmation check after each reminder
- Control/muting groups received parallel placebo (time management tips)

WTA elicitation: BDM mechanism

- Becker–DeGroot–Marschak (1964) mechanism
- Participants state minimum \$ to mute accounts for 4 weeks
- Fixed \$15 offer
- If stated WTA $\leq$ \$15: accepted into randomization
- If stated WTA $>$ \$15: offered chance to confirm at \$15
- Those declining \$15: completed survey, excluded from randomization
- Incentive compatibility: 1% of respondents in each of Waves 2 and 3 received a random offer $\sim U[\$0, \$50]$
  - If offer $\geq$ WTA: muting implemented + payment
  - If offer $<$ WTA: no muting, no payment
  - Excluded from analysis sample (per preregistration)